Day 07: Python Kurtosis Practical Blueprint

COMPLETE EXECUTED CODES & TECHNICAL LAB NOTES

Lab Manual Overview: Aaj ke practical session mein humne Kurtosis ke teeno structural mizaaj (Platykurtic, Leptokurtic, aur Mesokurtic) ko pehle Pure Manual Loops ke sath aur phir industrial libraries (SciPy & Pandas) ke sath step-by-step implement kiya hai. Niche aapke poore practical execution ke codes text formate me diye gaye hain taaki aap asani se copy aur practice kar sakein.

Section 1: Base Manual Setup & Standard Population Core Logic

Sabse pehle humne bina kisi library ke pure core Python loops ka use karke basic Mathematical formula ko implement kiya. Yeh code data ke deviation ka 4th power nikaalta hai aur standard sample scale ko mathematically capture karta hai.

```python
# 1. Hamaara Test Data aur Total Count (n)
data = [10, 12, 15, 18, 20, 22, 25, 30, 35]
n = len(data)

# 2. STEP 1: Mean (Average) Nikalna
total_sum = 0
for x in data:
    total_sum += x

mean_val = total_sum / n

# 3. STEP 2 & 3: Variance aur 4th Power ke liye Loops Chalana
sum_squared_deviation = 0
sum_fourth_deviation = 0

for x in data:
    # Har number se Mean ka faasla (Deviation)
    deviation = x - mean_val

    # Denominator (Neeche) ke liye: Square karke jodna
    sum_squared_deviation += deviation ** 2

    # Numerator (Upar) ke liye: Power 4 karke jodna
    sum_fourth_deviation += deviation ** 4

# 4. STEP 4: Final Variance aur Denominator (Sigma^4) ki Calculation
variance = sum_squared_deviation / n
denominator = variance ** 2  # Variance ka square hi sigma^4 hota hai

# 5. STEP 5: Numerator (4th Power ka Average)
numerator = sum_fourth_deviation / n

# 6. Final Kurtosis Score
kurtosis_score = numerator / denominator

# Result Print Karte Hain
print("============== MANUAL CODES RESULTS ==============")
print(f"Total Numbers (n)      : {n}")
print(f"Mean (Average)         : {mean_val:.2f}")
print(f"Numerator (4th Moment) : {numerator:.2f}")
print(f"Denominator (Sigma^4)  : {denominator:.2f}")
print(f"Final Kurtosis Score   : {kurtosis_score:.2f}")
print("==================================================")
```

Section 2: Industrial Short-cuts (SciPy & Pandas Integration)

Manual base built karne ke baad, humne professional analysts ka core weapon use kiya. SciPy ke through direct true population code and Pandas Series workflow ka direct shortcut execute kiya jahan Fisher scale validation ki poori comparison window milte hai.

```
import scipy.stats as stats

# Hamaara data
data = [10, 12, 15, 18, 20, 22, 25, 30, 35]

# Sirf ek line mein poora khel khatam!
scipy_kurtosis = stats.kurtosis(data, fisher=False)

print("============== SCIPY RESULT ==============")
print(f"SciPy Kurtosis Score: {scipy_kurtosis:.2f}")
print("==========================================")

import pandas as pd

# Data ko pandas ki series (column) mein convert kiya
df_column = pd.Series([10, 12, 15, 18, 20, 22, 25, 30, 35])

# Ek line mein shortcut command
pandas_kurtosis = df_column.kurt()

print("============== PANDAS RESULT ==============")
print(f"Pandas Sample Kurtosis: {pandas_kurtosis:.2f}")
print("===========================================")
```

Section 3: Type 2 Mizaaj - Leptokurtic Outlier Risk Validation

Yahan humne data ke andar ek extreme sudden spike (Outlier value: 110) add kiya. Is heavy shock ki wajah se data ki peak aur tails ka weight badha, jise humne manual loop aur short code dono se mathematically confirm kiya.

3.1 Leptokurtic Pure Manual Loop Execution

```
# 1. New Dataset (Leptokurtic - Outlier Waala Data)
lepto_data = [10, 10, 10, 10, 110]
n = len(lepto_data)

# 2. Mean Calculation
total_sum = 0
for x in lepto_data:
    total_sum += x
mean_val = total_sum / n  # Mean aayega 30.00

# 3. Deviation, Squares (Denominator) aur Power 4 (Numerator) Loop
sum_squared_deviation = 0
sum_fourth_deviation = 0

for x in lepto_data:
    deviation = x - mean_val

    # Square for Denominator
    sum_squared_deviation += deviation ** 2

    # Power 4 for Numerator
    sum_fourth_deviation += deviation ** 4

# 4. Final Calculations
variance = sum_squared_deviation / n      # Variance = 1600.00
denominator = variance ** 2               # Sigma^4 = 2,560,000.00

numerator = sum_fourth_deviation / n       # 4th Moment = 8,320,000.00

# 5. Final Kurtosis Score
kurtosis_score = numerator / denominator

print("============== LEPTOKURTIC MANUAL RESULT ==============")
print(f"Mean (Average)         : {mean_val:.2f}")
print(f"Variance (Sigma^2)     : {variance:.2f}")
print(f"Numerator (4th Moment) : {numerator:.2f}")
print(f"Denominator (Sigma^4)  : {denominator:.2f}")
print(f"Final Kurtosis Score   : {kurtosis_score:.2f}")
print("=======================================================")
```

3.2 Leptokurtic SciPy One-Line Shortcut

```
import scipy.stats as stats

# Hamaara outlier waala data
lepto_data = [10, 10, 10, 10, 110]

# Sirf ek line mein shock-absorber shortcut!
scipy_lepto_score = stats.kurtosis(lepto_data, fisher=False)

print("============== SCIPY LEPTO RESULT ==============")
print(f"SciPy Leptokurtic Score: {scipy_lepto_score:.2f}")
print("================================================")
```

Section 4: Type 3 Mizaaj - Mesokurtic Perfect Symmetrical Trend

Is pattern mein humne mathematical symmetry ke rules ke hisab se ek completely balanced standard curve baseline tayyar ki, jahan score hamesha nominal default benchmark (3.00) ke bilkul close aakar stabilize hota hai.

4.1 Mesokurtic Pure Manual Loop Execution

```
# 1. Symmetrical Dataset (Mesokurtic Trend)
meso_data = [10, 14, 16, 18, 20, 20, 22, 24, 26, 30]
n = len(meso_data)

# 2. Mean Calculation
total_sum = 0
for x in meso_data:
    total_sum += x
mean_val = total_sum / n  # Mean aayega ekdum perfect 20.00

# 3. Deviation, Squares (Denominator) aur Power 4 (Numerator) Loop
sum_squared_deviation = 0
sum_fourth_deviation = 0

for x in meso_data:
    deviation = x - mean_val

    # Square for Denominator
    sum_squared_deviation += deviation ** 2

    # Power 4 for Numerator
    sum_fourth_deviation += deviation ** 4

# 4. Final Calculations
variance = sum_squared_deviation / n      # Variance = 31.20
denominator = variance ** 2               # Sigma^4 = 973.44

numerator = sum_fourth_deviation / n       # 4th Moment = 2658.24

# 5. Final Kurtosis Score
kurtosis_score = numerator / denominator

print("============== MESOKURTIC MANUAL RESULT ==============")
print(f"Mean (Average)         : {mean_val:.2f}")
print(f"Variance (Sigma^2)     : {variance:.2f}")
print(f"Numerator (4th Moment) : {numerator:.2f}")
print(f"Denominator (Sigma^4)  : {denominator:.2f}")
print(f"Final Kurtosis Score   : {kurtosis_score:.2f}")
print("======================================================")
```

4.2 Mesokurtic SciPy One-Line Shortcut

```
import scipy.stats as stats

# Hamaara symmetrical balanced data
meso_data = [10, 14, 16, 18, 20, 20, 22, 24, 26, 30]

# Ek line mein direct baseline score!
scipy_meso_score = stats.kurtosis(meso_data, fisher=False)

print("============== SCIPY MESO RESULT ==============")
print(f"SciPy Mesokurtic Score: {scipy_meso_score:.2f}")
print("===============================================")
```

Section 5: Type 1 Mizaaj - Platykurtic Smooth Flat Pattern

Aakhiri step mein humne hamara pehla original dataset implement kiya jo upar se ekdum sapat (flat) hota hai aur isme extreme tail risk zero hoti hai. Iska score hamesha standard value 3 se chhota milta hai.

5.1 Platykurtic Comprehensive Code (Manual + Short Combined)

```python
import scipy.stats as stats

# =====================================================================
# PART 1: PURE MANUAL LOOP CODE (Zero Library Way)
# =====================================================================

# 1. Hamaara Sapat / Flat Data (Platykurtic Trend)
platy_data = [10, 12, 15, 18, 20, 22, 25, 30, 35]
n = len(platy_data)

# 2. Mean Calculation
total_sum = 0
for x in platy_data:
    total_sum += x
mean_val = total_sum / n  # Mean = 20.78

# 3. Deviation, Squares (Denominator) aur Power 4 (Numerator) Loop
sum_squared_deviation = 0
sum_fourth_deviation = 0

for x in platy_data:
    deviation = x - mean_val

    # Square for Denominator
    sum_squared_deviation += deviation ** 2

    # Power 4 for Numerator
    sum_fourth_deviation += deviation ** 4

# 4. Final Calculations
variance = sum_squared_deviation / n      # Variance = 60.17
denominator = variance ** 2               # Sigma^4 = 3620.43

numerator = sum_fourth_deviation / n       # 4th Moment = 7673.03

# 5. Final Kurtosis Score
manual_platy_score = numerator / denominator


# =====================================================================
# PART 2: SHORT CODE (SciPy Industry Shortcut Way)
# =====================================================================

# Ek line mein direct population standard score!
scipy_platy_score = stats.kurtosis(platy_data, fisher=False)
```

```
# ======================================================================
# FINAL RESULTS DISPLAY
# ======================================================================
print("============== PLATYKURTIC COMPLETE RESULTS ==============")
print(f"Mean (Average)          : {mean_val:.2f}")
print(f"Manual Loop Score       : {manual_platy_score:.2f}")
print(f"SciPy Shortcut Score    : {scipy_platy_score:.2f}")
print("==========================================================")
```

5.2 Platykurtic Dedicated Short Code Verification

```
import scipy.stats as stats

# Hamaara sapat (flat) data
platy_data = [10, 12, 15, 18, 20, 22, 25, 30, 35]

# Ek line mein direct short code!
scipy_platy_score = stats.kurtosis(platy_data, fisher=False)

print("============== SCIPY PLATY RESULT ==============")
print(f"SciPy Platykurtic Score: {scipy_platy_score:.2f}")
print("================================================")
```